

Professional Communications & Research Methods - NUIST

Engineering Technology

May, 2026

Professional Communications & Research Methods - NUIST (A35617)

Short Title:	Prof Comm & Resrch Methods (NUIST)	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Engineering Practise	
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Credits	5	Level: Postgraduate

Description of Module / Aims

The Professional Communications and Research Methods module is designed to develop student knowledge, skills and competencies in professional communications and research methods including, project management, research methods (quantitative and qualitative), technical presentations and writing skills, health and safety and ethics in engineering. The student will attain skills in critical assessment of high end technologies from secondary research sources and their integration towards technology convergence.

Programmes

RESA-0162	MEng in Electronic Engineering (WD_EELEN_R)	1 / 1 / M
	MEng in Electronic Information Systems (WD_EELEN_RI)	1 / 2 / M
RESA-0162	MSc in Innovative Technology Engineering (WD_CINNT_R)	1 / 1 / M
RESA-0162	PG Dip in Engineering in Electronic Engineering (WD_EELEN_G)	1 / 1 / M
RESA-0162	PG Dip in Science in Innovative Technology Engineering (WD_CINNT_G)	1 / 1 / M

Indicative Content

- Project Management Skills: Project planning incorporating time and resource management
- Technical writing and Presentation Skills
- Secondary Research Techniques including patent searching, academic journal searching and Boolean search strategies
- Quantitative and Qualitative Research Methods
- Critical analysis of experimental data using SPSS and associated techniques
- Introduction to Intellectual Property Rights

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Write technical reports, communicating a balanced review of complex and/or controversial ideas; demonstrate appropriate academic and professional standards of written communication, including spelling, grammar, structure, citation, and presentation in both electronic and hardcopy forms.
2. Develop an engineering project plan and to effectively manage time/resources when conducting a project.
3. Conduct secondary research and critically evaluate secondary information sources.
4. Use statistical analysis software.
5. Effectively communicate to both technical and non-technical audiences.
6. Present and defend a detailed and feasible research proposal on a specific topic of interest.
7. Critically evaluate the research designs and techniques identified for a research project.

Learning and Teaching Methods

- Statistical procedures for methodology and sampling
- Guest Lectures on selected topics common with module themes from the programme
- Guide to effective technical writing and academic journal submissions
- Case studies for critical analysis
- Communications workshops
- Presentation skills

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Reflective Journal</i>	40%	1,2,3,5,6
<i>Essay</i>	30%	1,3,4,5,7
<i>Presentation</i>	30%	3,5,6

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	12	
Independent Learning	99	